

Pierre BELAMRI

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SUMMARY

PhD candidate (graduating 2026) at Mines Paris – PSL / CNRS, specializing in **self-supervised and multimodal machine learning** for scientific data. Research focuses on latent representation learning over microstructure and mechanical testing data (EBSD, nanoindentation, acoustic emission) for sub-grain elastoplasticity inference. Visiting researcher at UIUC and UCSB. Lecturer on Transformers at Mines Paris. Founder of an AI SaaS company (Fortune 500 clients). All research open-sourced.

WORK EXPERIENCE

PhD Researcher — CNRS & Mines Paris (PSL), Paris Jan 2024 – Present

- **Thesis:** *ML for anomaly detection in nanoindentation to identify crystal elastoplasticity at the sub-grain level*. Supervisors: D. Ryckelynck, D. Texier.
- Developed a self-supervised pipeline to learn structured latent representations of high-dimensional microstructure data (EBSD orientation maps, P-h curves, acoustic waveforms).
- Designed a multimodal data registration pipeline coupling EBSD, nanoindentation, and acoustic emission into a unified dataset (oxygen-enriched titanium); manuscript in preparation.
- Pre-trained a Quaternion Vision Transformer on 2M synthetic EBSD patches (128×128×4 px); **first-author**, *Materials & Design* (2025). Presented at **TMS AIM2025**.
- Jean-Zay HPC (IDRIS); largest runs: 8×A100 and 4×H100. All code and datasets open-sourced.

Visiting PhD Student — University of Illinois Urbana-Champaign (UIUC) Oct 2025 – Nov 2025

- Visiting scholarship in the group of Prof. J.-C. Stinville (Materials Science & Engineering).
- **2nd place / 77 international teams + Polaron Sponsor Prize**, 2025 AI/ML for Microscopy Hackathon: developed an ML pipeline for automated EBSD acquisition directing spatial sampling toward out-of-distribution regions via a Mahalanobis-distance latent-space prior, significantly reducing electron beam exposure. [Submission](#)

Visiting PhD Student — UC Santa Barbara (UCSB) May 2025 – Jun 2025

- Visiting scholarship in the group of Prof. Sam Daly (Mechanical Engineering).
- Integrated acoustic emission sensing into micro-mechanical testing frameworks; engineered latent features from AE waveforms for multimodal ML architectures.
- Facilitated strategic alignment for a joint initiative between Mines Paris, UCSB, NASA, and Synchrotron SOLEIL on advanced material characterization.

Lecturer — Mines Paris – PSL, Paris 2024 – Present

Teaching the **Transformers** course to MSc engineering students: architecture, attention mechanisms, pre-training, and fine-tuning.

Founder — VH Systems / EngageAI, Paris May 2024 – Present

Built and commercialized **EngageAI**, an AI-powered OSINT stakeholder intelligence platform (web search agents, GCP, ReactJS). 4 active clients including 3 Fortune 500 companies (Life Science, Pharma, Telecommunications). Fully bootstrapped.

Machine Learning Consultant — University of Zurich Apr 2024 – Jun 2025

- Fine-tuned a representation head on **NV-Embed-v2** on a self-curated 2024 US election corpus; the latent space exhibited spontaneous separation along the political compass with no a priori labels.
- Set up multi-GPU training infrastructure; advised on hyperparameter tuning and data preprocessing for political science researchers.

ML Engineer — Consultys, Lausanne Mar 2023 – Nov 2023

Deployed open-source LLMs on client cloud instances (Docker). Built a modular multitask conversational agent for Research Scientists. Master's thesis on **LLM quantization** (GPTQ and variants).

ML Intern (BI) — Amazon, Berlin Mar 2022 – Jul 2022

Trained a competitor job-posting NLP classifier; deployed end-to-end on AWS (CDK, Docker, ETL) with cost-efficient ARM inference.

Software Engineer Intern — PMI, Lausanne Sep 2021 – Feb 2022

Selenium scrapers on Tor for illicit trade monitoring; graph network and dark web intelligence for tobacco contraband prevention.

Research Intern — CAOR Mines Paris, Paris Mar 2021 – Apr 2021

CV + Deep RL for robot arm manipulation: **98.3% success rate** in under 30k training epochs. [Preprint](#).

PROJECTS

CRAFT-JEPA (independent research)

GitHub

Fine-tuning **V-JEPA** on the VPT dataset (4.5 TB Minecraft gameplay + controls) to probe long-horizon planning in a stochastic environment. First run converged; architecture evaluation in progress. Trained on 4×H100 (self-funded).

EDUCATION

2024 – 2026 **PhD** in Applied Machine Learning for Materials Science — **Mines Paris – PSL** / CNRS
2019 – 2023 **MSc & Diplôme d’Ingénieur** (Executive Engineering) — **Mines Paris – PSL**
2017 – 2019 Classes préparatoires — **Lycée Marcel Sembat**

SKILLS

Research	Self-supervised learning, multimodal representation learning, anomaly detection, transformers, CNNs, RL, LLM quantization, NLP
ML Stack	PyTorch, TensorFlow, HuggingFace, scikit-learn SLURM/HPC (Jean-Zay), distributed training (8×A100, 4×H100), AWS, GCP, Docker
Engineering	Python, JavaScript/ReactJS, SQL, C#/Unity
Languages	French (native), English (fluent)

ACHIEVEMENTS

Competitive Gaming Global rank **14th** in competitive Minecraft PvP (Kohi, 2016); core member of *VorHerr*, ranked world #1 PvP team at the time.

PUBLICATIONS

Belamri, Pierre et al. (May 2021). “Towards a Sample Efficient Reinforcement Learning Pipeline for Vision Based Robotics”. In: URL: <https://arxiv.org/pdf/2105.09719.pdf>.
Belamri, Pierre, Henry Proudhon, et al. (2025). “Quaternion-Based Vision-Transformer for Polycrystalline EBSD Scans Pre-Trained on Large-Scale Synthetic Data”. In: *Materials & Design*, p. 114599.

Last updated: October 8, 2026